Digital Signal Processing

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Design of linear phase FIR filters using the frequency sampling method



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at Beijing Jiaotong University

- Design principle
- Design method
- Method improvement

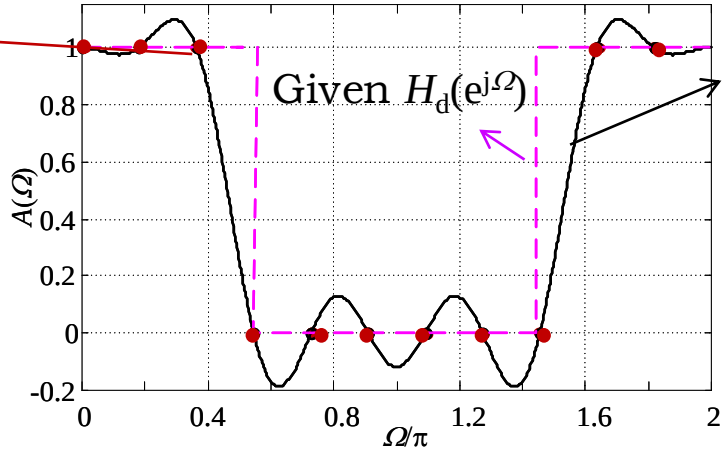
Design principle

Window function method: based on approximation in the time domain

Frequency sampling method: based on approximation in the frequency domain

Make the frequency response $H(e^{j\Omega})$ of the designed M-order FIR filter equal to the frequency response of the given digital filter at M+1 sampling points $\{\Omega_m: m=0,1,\ldots, M\}$

M+1 sampling points



The designed $H(e^{j\Omega})$ of the M-order FIR DF

$$H(e^{j\Omega_m}) = H_d(e^{j\Omega_m})$$

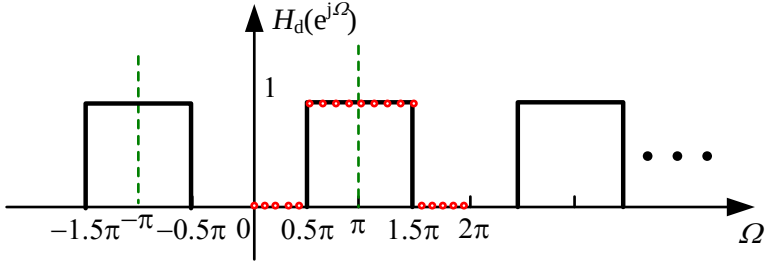
$$\Omega_m = \frac{2\pi}{M+1}m, \quad m = 0,1,\cdots,M$$

Design principle

The goal of FIR digital filter design is to determine its unit impulse response $h[k]$

How to find $h[k]$ that satisfies $H(e^{j\Omega_m}) = H_d(e^{j\Omega_m})$ and has a linear phase filter

Because $H_d(e^{j\Omega}) = \sum_{k=0}^{M} h[k]e^{-jk\Omega}$



If $\Omega_m = \frac{2\pi}{M+1}m, \quad m = 0,1,\cdots,M$

M+1 point DFT

Then $H_d(e^{j\Omega_m}) = H_d[m] = \sum_{k=0}^{M} h[k]W_{M+1}^{mk}$

$W_{M+1} = e^{-j\frac{2\pi}{M+1}}$

Design principle

How to find $h[k]$ that satisfies $H(e^{j\Omega_m}) = H_d(e^{j\Omega_m})$ and has a linear phase filter?

$$H_d[m] = \sum_{k=0}^{M} h[k] W_{M+1}^{mk} = \mathrm{DFT}\{h[k]\} \Longrightarrow h[k] = \mathrm{IDFT}\{H_d[m]\}$$

To ensure that the designed filter has linear phase, then

$$H_d(e^{j\Omega}) = e^{j(-0.5M\Omega+\beta)} A_d(\Omega) \qquad (\beta = 0 \text{ or } \pi/2)$$

Namely $H_d[m] = H_d(e^{j\Omega})\Big|_{\Omega=\frac{2\pi}{M+1}m} = e^{j\beta} e^{-j\frac{M}{M+1}m\pi} A_d\left(\frac{2\pi m}{M+1}\right)$ $A_d[m]$

Design method

1.The type and amplitude function $A_d(\Omega)$ of FIR DF are determined by $H_d(e^{j\Omega})$

2.Determine the phase $\varphi_d(\Omega)$ of the linear phase FIR filter based on the type $\varphi_d(\Omega)$

$$\varphi_d(\Omega) = -0.5M\Omega + \beta \qquad (\beta = 0 \text{ or } \pi/2)$$

3.Determine the values $H_d[m]$ of $A_d(\Omega)e^{j\varphi_d(\Omega)}$ at $M+1$ sampling points within the interval $\Omega \in [0, 2\pi)$

$$H_d[m] = e^{j\beta} e^{-j\frac{M\pi}{M+1}m} A_d\left(\frac{2\pi}{M+1}m\right)$$

4.Perform an M+1 point IDFT on $H_d[m]$ to obtain a finite-length causal sequence $h[k]$

$$h[k] = \text{IDFT}(H_d[m])$$

Example: design a linear phase FIR filter using the frequency sampling method, where the amplitude response approximates a high-pass filter $H_d(e^{j\Omega})$ with a cut-off frequency $\Omega_c=5\pi/7$ rad, the order $M=6$.

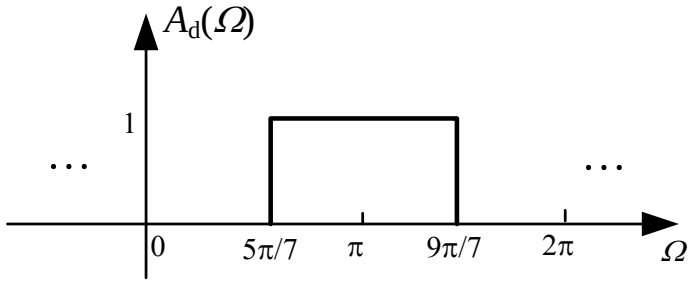
Solution:

(1) Determine the type of linear phase FIR filter based on $H_d(e^{j\Omega})$:

The high-pass filter can be of Type I or Type IV. For this problem, Type I is selected.

Determine the amplitude function $A_d(\Omega)$ based on $H_d(e^{j\Omega})$:

$$A_d(\Omega)=\begin{cases}1, & 5\pi/7\le|\Omega|\le\pi\\ 0 & \text{others}\end{cases}$$



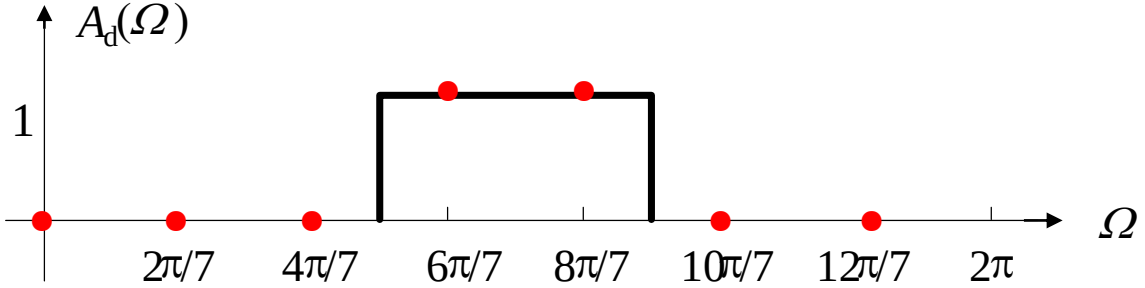
(2) Determine the phase $\varphi_d(\Omega)$ of the linear phase FIR filter based on the type:

Type I linear phase $\varphi_d(\Omega)$ $\quad\varphi_d(\Omega)=-0.5M\Omega=-3\Omega$

Example: design a linear phase FIR filter using the frequency sampling method, where the amplitude response approximates a high-pass filter $H_c(e^{j\Omega})$ with a cut-off frequency $\Omega_c=5\pi/7$ rad, the order $M=6$.

(3) Determine the values $H_d[m]$ of $H_d(e^{j\Omega})$ at 6+1=7 sampling points within the interval $\Omega\in[0,2\pi)$

$$H_d[m]=e^{-j3\Omega}A_d(\Omega)\Big|_{\Omega=\frac{2\pi}{7}m}=e^{-j\frac{6\pi}{7}m}A_d[m]=[0,\ 0,\ 0,\ e^{-j18\pi/7},e^{-j24\pi/7},\ 0,\ 0]$$



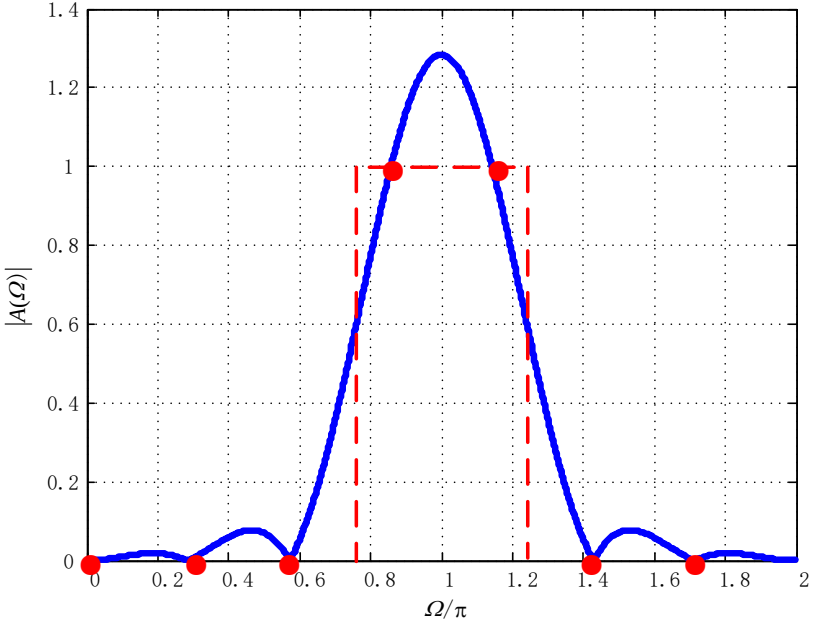
(4) Obtain the unit impulse sequence:

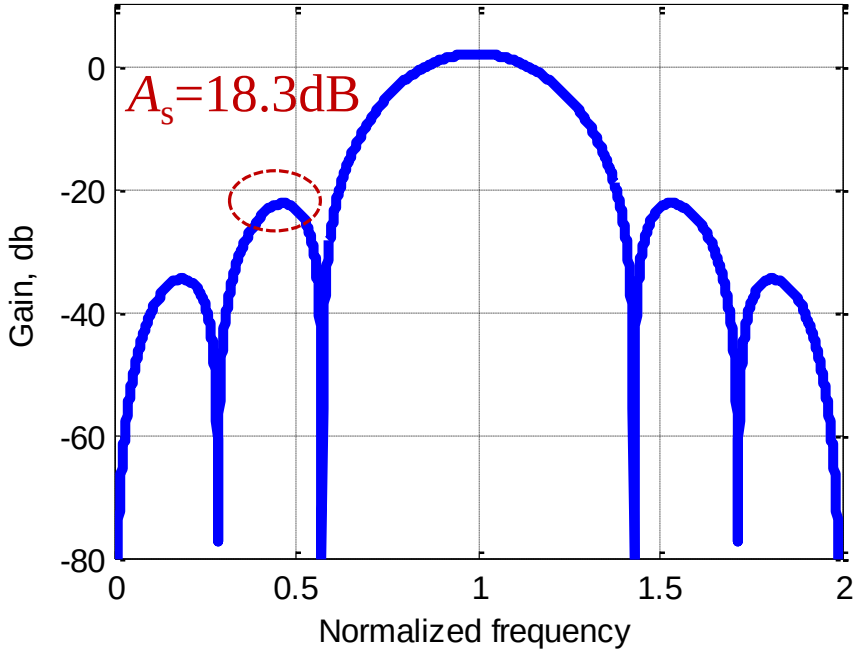
$$h[k]=\mathrm{IDFT}(H_d[m])=\frac{2}{7}\cos\left[\frac{6\pi}{7}[k-3]\right]$$

Example: design a linear phase FIR filter using the frequency sampling method, where the amplitude response approximates a high-pass filter $H_c(e^{j\Omega})$ with a cut-off frequency $\Omega_c=5\pi/7$ rad, the order $M=6$.

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Design results $M=6$, $A_s=22.2$dB



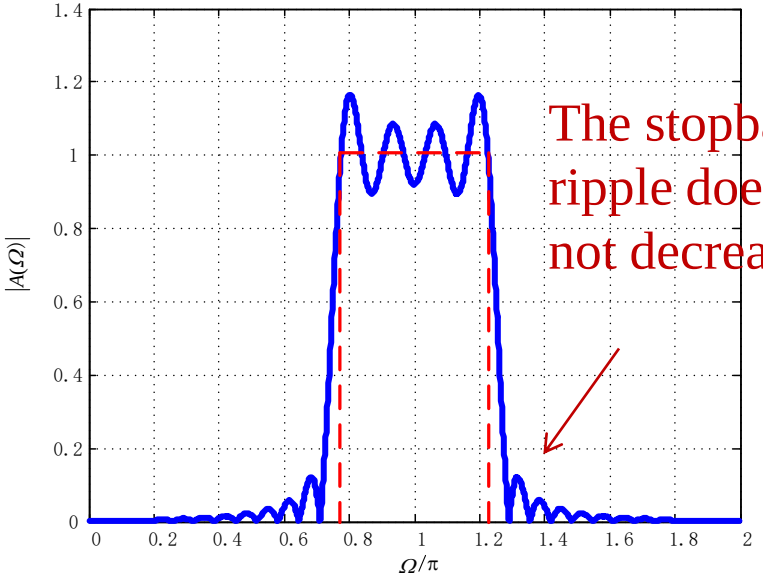


For example, design a linear phase FIR filter using the frequency sampling method, where the amplitude response approximates a high-pass filter $H_d(e^{j\Omega})$ with a cut-off frequency $\Omega_c=5\pi/7$ rad, the order $M=6$.

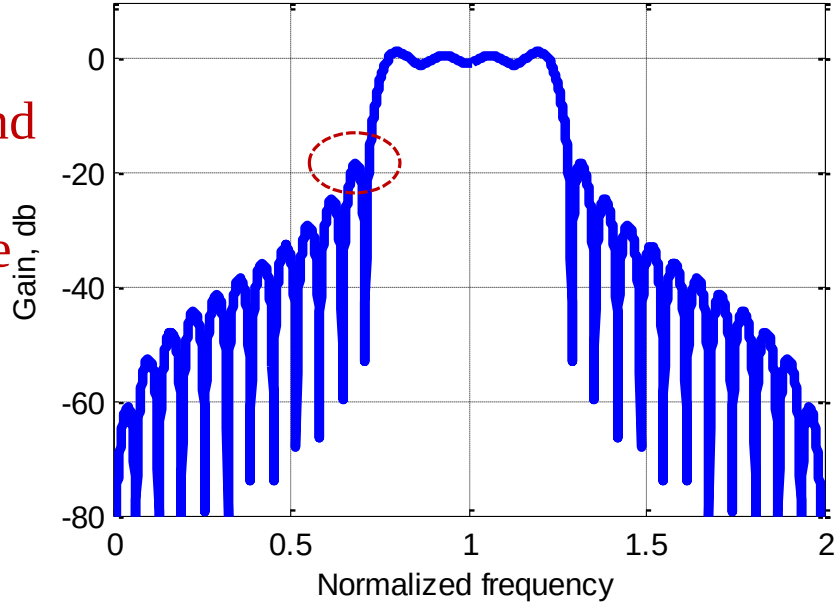
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Design results Increase the order of the filter and set $M=30$,

$A_s=18.3$dB, The stopband attenuation does not improve



The stopband ripple does not decrease



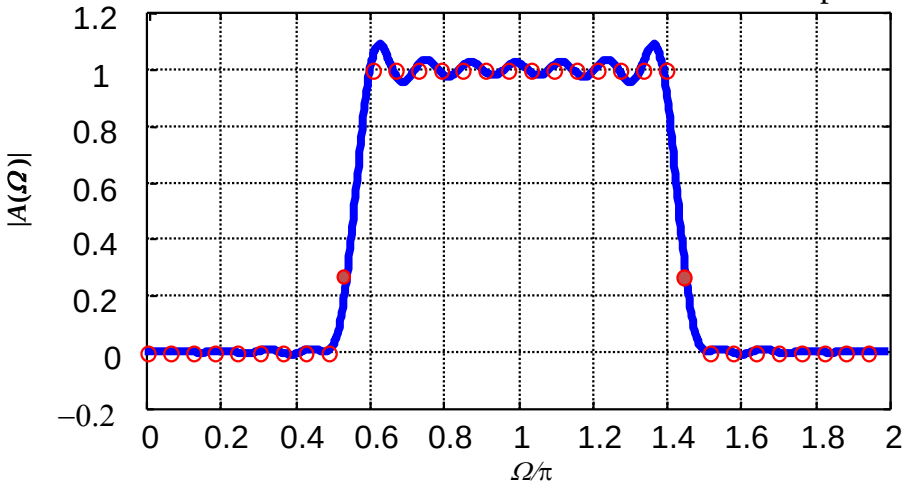
How can the stopband attenuation be improved?

Method improvement

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Cause analysis: The reason why increasing the order of the filter cannot improve the stopband attenuation of the designed filter is that there is a jump from 0 to 1 at the given sample points.

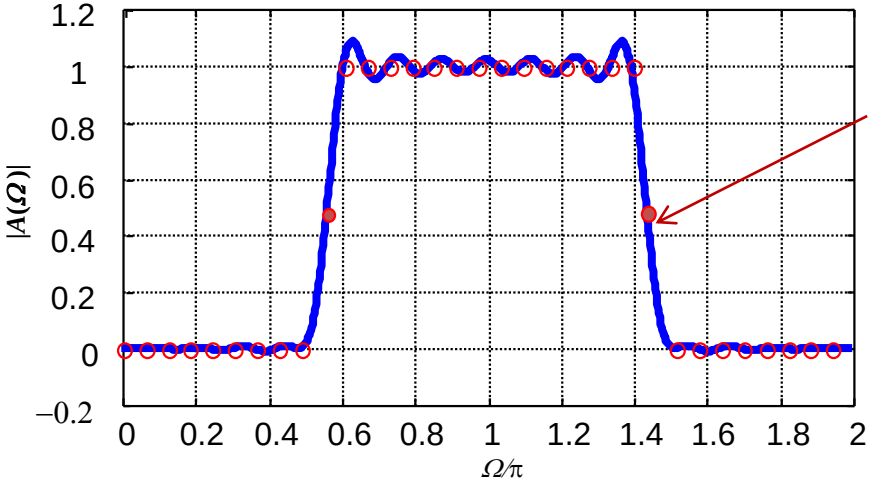
Improvement methods: To improve the amplitude characteristic of the filter and increase the stopband attenuation, transition points with amplitudes between 0 and 1 can be set within the transition band [Ω_p, Ω_s].



Method improvement

For example, design a linear phase FIR filter using the frequency sampling method, where the amplitude response approximates a high-pass filter $H_d(e^{j\Omega})$ with a cut-off frequency $\Omega_c=5\pi/7$ rad, the order $M=30$. Try to improve the stopband attenuation by setting transition points with amplitudes between 0 and 1 within the transition band $[\Omega_p, \Omega_s]$

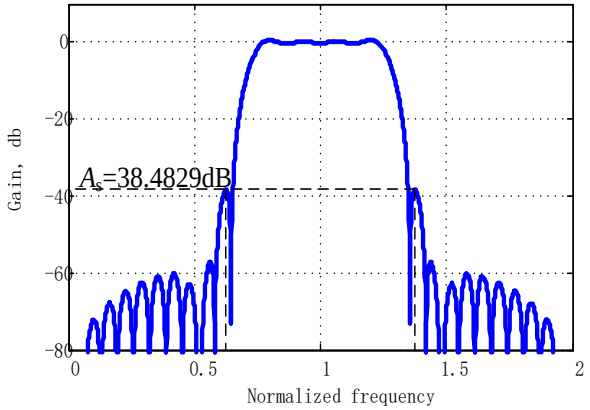
Set one transition point within the transition band $[\Omega_p, \Omega_s]$



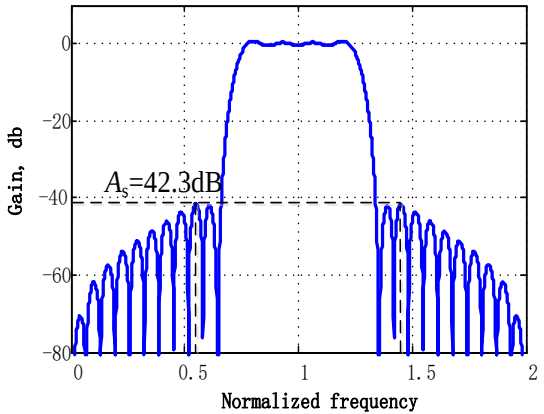
Assign amplitude values separately for 0.38, 0.28, 0.18

Method improvement

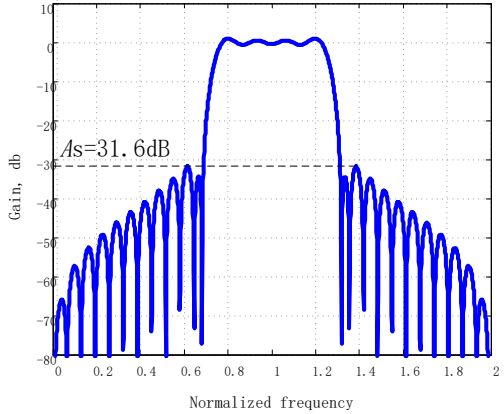
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Transition point value 0.38



Transition point value 0.28



Transition point value 0.18

Adding transition points can reduce the sidelobe fluctuations. When the transition point value is 0.28, the sidelobe fluctuations are relatively small.

Using MATLAB to Realize the frequency sampling method

1. Determine the values $H_d[m]$ at the sampling points based on the filter's frequency response.

2. Performing an M+1 point IDFT on $H_d[m]$ yields $h[k]$.

transition point → Set within the transition band $[\Omega_p, \Omega_s]$ → Amplitude value of transition point.

Example: design a Type I linear phase high-pass digital filter with Ω_p=0.6π rad using the frequency sampling method.

Analysis: Frequency sampling method No transition point

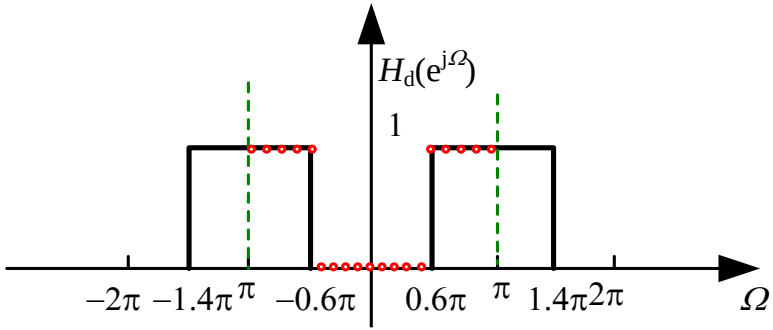
Key function:

Calculate the frequency corresponding to the sampling point

Wm=2*pi*m./(M+1);

Set as high pass

Ad=double([Wm>=Wp]);



Determine the value of the filter at the sampling points

Hd=Ad.*exp(-j*0.5*M*Wm);

Hd=[Hd conj(fliplr(Hd(2:end)))]; Determine the values within the range [*M*/2,*M*] base on the conjugate symmetry characteristic of Hd

Key parameters:

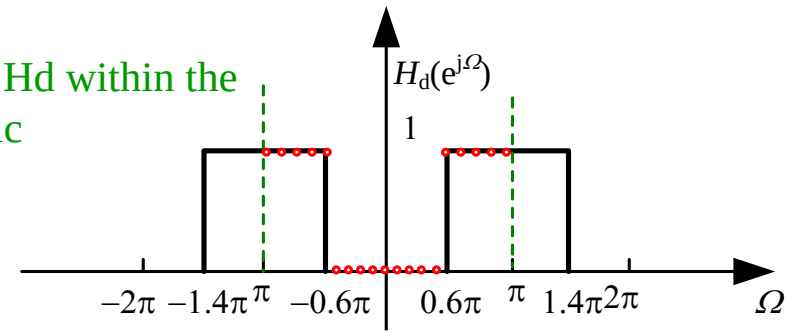
M=32;Wp=0.6*pi; m=0:M/2; *M* is the order of the filter, and *m* represents the sampling points within the passband

Example: design a Type I linear phase high-pass digital filter with Ω_p=0.6π rad using the frequency sampling method.

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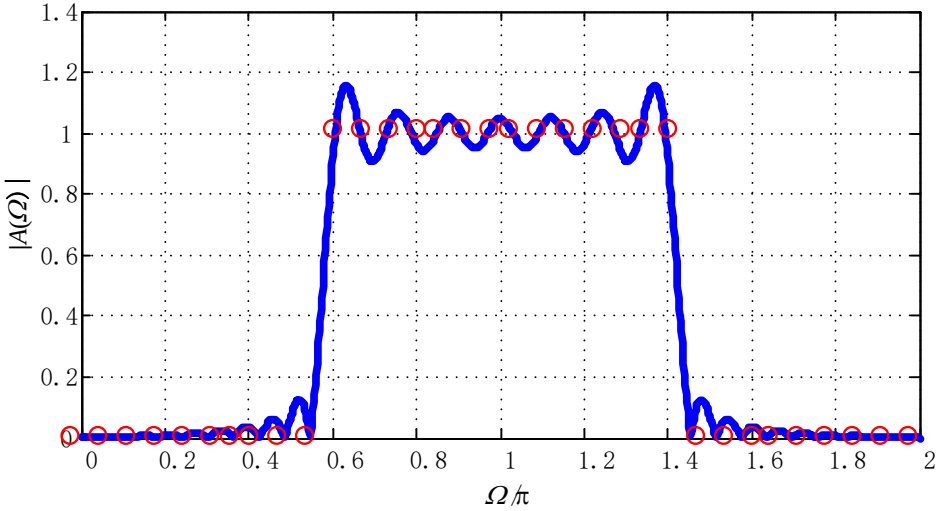
Solution:

```
%Design a Type I high-pass filter using the frequency sampling method
M=32;Wp=0.6*pi; m=0:M/2; %Define filter parameters, where m represents
the sampling points within the passband
Wm=2*pi*m./(M+1); %Frequencies corresponding to the sampling points
Ad=double([Wm>=Wp]); %set as high pass
Hd=Ad.*exp(-j*0.5*M*Wm); %Determine the values of the filter at the
sampling points
Hd=[Hd conj(fliplr(Hd(2:end)))]; %Determine the values of Hd within the
range [M/2,M] based on its conjugate symmetry characteristic
h=real(ifft(Hd)); %Perform an inverse discrete Fourier
transform (IDFT)and take the real part
w=linspace(0.1,pi,1000);
H=freqz(h,[1],w); %Calculate the frequency response
plot(w/pi,(abs(H)),'linewidth',3);grid; %Plot the magnitude response
xlabel('\Omega/\pi');ylabel('A(\Omega)');
figure;plot(w/pi,20*log10(abs(H)),'linewidth',3);grid; %Plot the gain response
xlabel('Normalized frequency');  ylabel('Gain, dB');
```

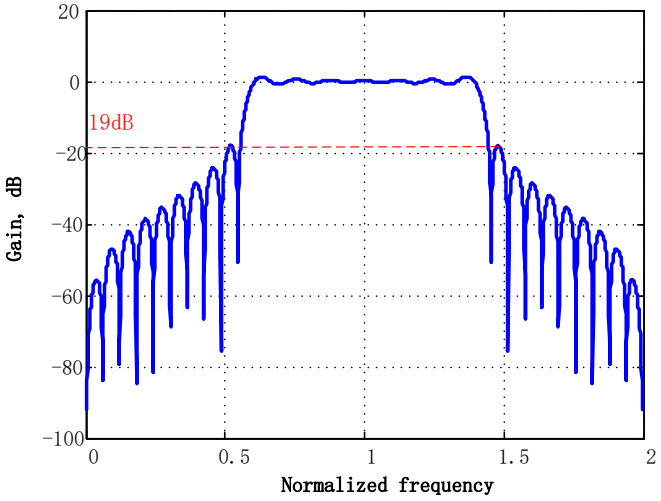


Sampling point illustration

Example: design a Type I linear phase high-pass digital filter with Ω_p=0.6π rad using the frequency sampling method.



Magnitude response

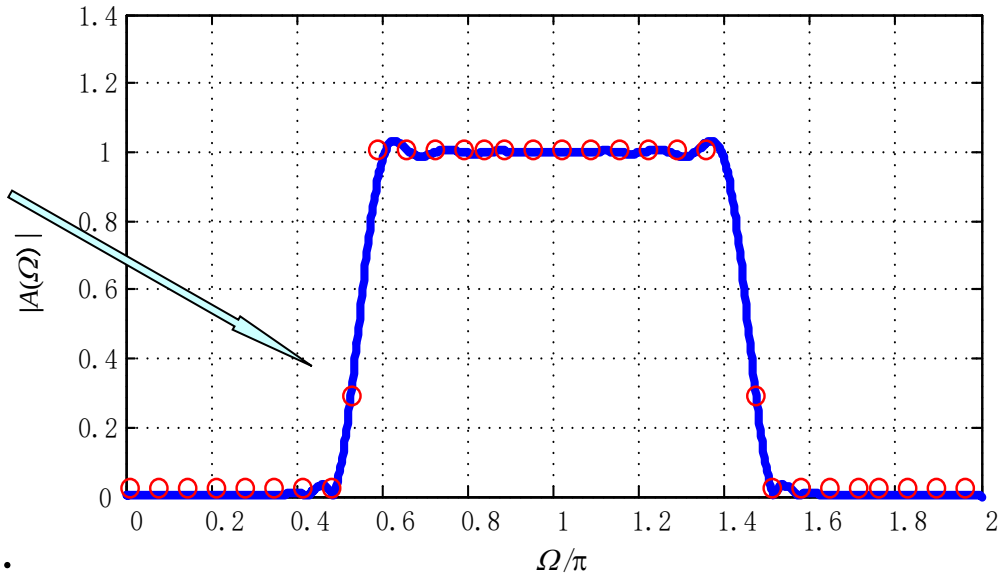


Gain response

M=32 A_p=0.8473dB, A_s=19dB

Example: design a Type I linear phase high-pass digital filter with Ω_p=0.6π rad using the frequency sampling method.

Design a high-pass filter amplitude function using the frequency sampling method after **setting** one transition point.



Add two statements:

```
mtr=ceil(Wp*(M+1)/(2*pi));
Ad(mtr)=0.28;    %Set the transition point value to 0.28
```

Example: design a Type I linear phase high-pass digital filter with Ω_p=0.6π rad using the frequency sampling method.

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Analysis: Frequency sampling method With transition point

Key function:

Calculate the frequency corresponding to the sampling points

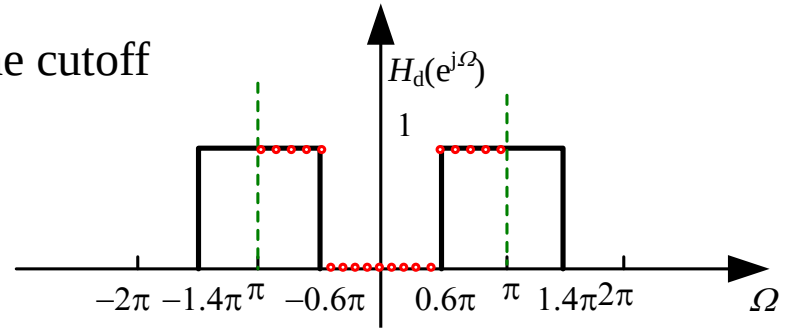
Wm=2*pi*m./(M+1);

Calculate the sampling points corresponding to the cutoff frequency of the passband, and round up

mtr=ceil(Wp*(M+1)/(2*pi));

Set as high pass

Ad=double([Wm>=Wp]);



Ad(mtr)=0.28; Select an appropriate transition point value based on experiments

Determine the values of the filter at the sampling points

Hd=Ad.*exp(-j*0.5*M*Wm);

Hd=[Hd conj(fliplr(Hd(2:end)))]; Determine the values within the range [*M*/2,*M*] base on the conjugate symmetry characteristic of Hd

Key parameters:

M=32;Wp=0.6*pi; m=0:M/2; *M* is the order of the filter, and *m* represents the sampling points within the passband

Example: design a Type I linear phase high-pass digital filter with Ω_p=0.6π rad using the frequency sampling method.

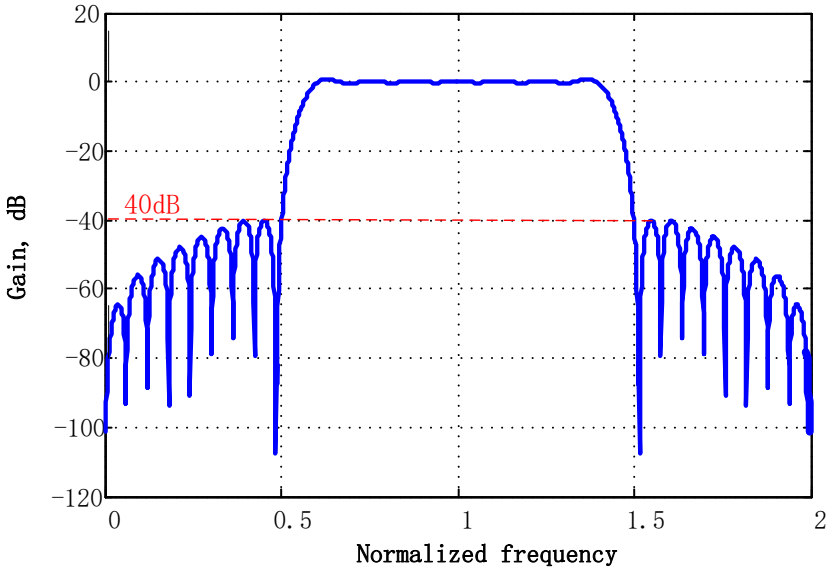
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Solution:

```
M=32;Wp=0.6*pi; m=0:M/2; %Defining filter parameters, where m represents the sampling points in the passband
Wm=2*pi*m./(M+1); %The frequency corresponding to the sampling points
mtr=ceil(Wp*(M+1)/(2*pi)); %Calculate the sampling points corresponding to the cutoff frequency of the passband, and round up
Ad=double([Wm>=Wp]);%Set as high pass
Ad(mtr)=0.28;   %Set the transition point
Hd=Ad.*exp(-j*0.5*M*Wm); %Determine the values of the filter at the sampling points
Hd=[Hd conj(fliplr(Hd(2:end)))]; %Determine the values within the range [M/2,M] base on the conjugate symmetry characteristic of Hd
h=real(ifft(Hd)); %Perform an inverse discrete Fourier transform and take the real part
w=linspace(0.1,pi,1000);
H=freqz(h,[1],w); %Calculate the frequency response
plot(w/pi,(abs(H)), 'linewidth',3);grid; % Plot the magnitude response
xlabel('\Omega/\pi');ylabel('A(\Omega)');
figure;plot(w/pi,20*log10(abs(H)), 'linewidth',3);grid; % Plot the gain response
xlabel('Normalized frequency');  ylabel('Gain, dB');
```

Setting up a single transition point

Example: design a Type I linear phase high-pass digital filter with $\Omega_p=0.6\pi$ rad using the frequency sampling method.



$M=32,\ A_p=0.43\text{dB},\ A_s=40\text{dB}$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.